

A PROJECT REPORT ON

Grammatical Error Correction in Marathi

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ABSTRACT

Marathi, a morphologically rich language, lacks dedicated research in Grammatical Error Correction (GEC) despite advancements in NLP for other languages. This gap exists primarily due to the absence of publicly available datasets and benchmarks for Marathi GEC. To bridge this gap, we conducted a comprehensive survey of existing Marathi NLP research, analyzing datasets, language models, and linguistic challenges. Additionally, we reviewed GEC efforts in other languages to identify effective methodologies and state-of-the-art models applicable to Marathi. Based on these insights, we systematically constructed a high-quality dataset of incorrect-correct sentence pairs, with an initial focus on fundamental Subject-Object-Verb (SOV) sentence structures. We fine-tuned three transformer-based models—IndicBART, Marathi-T5, and Varta-T5—on this dataset to assess their effectiveness in Marathi GEC. Performance evaluation using the BLEU metric indicates that IndicBART and Varta-T5 significantly outperform Marathi-T5, achieving a BLEU score of 0.98. This research establishes a foundation for Marathi GEC by providing both a benchmark dataset and strong baseline models, contributing to the broader development of Marathi NLP applications.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Our project focuses on developing a Grammatical Error Correction (GEC) system for Marathi, leveraging the latest advancements in Natural Language Processing (NLP) for Indic languages. Foundational resources such as MahaCorpus, MahaBERT, MahaNER, MahaGPT, and MahaFT provide essential datasets and pre-trained models that enable fine-tuning for error detection and correction tasks.

We explore a variety of deep learning models, including CNNs, LSTMs, transformers, and Seq2Seq architectures, for identifying and correcting grammatical errors at both the sentence and paragraph levels. Special attention is given to handling code-mixed texts (Marathi-English), named entity recognition (NER), and gender agreement, which are crucial for maintaining grammatical consistency.

Additionally, techniques from Neural Machine Translation (NMT) and synthetic error generation methods are adapted to augment datasets and improve model performance. Models like MuRIL, mT5, Varta-T5, and approaches from Hindi GEC systems are also adapted for Marathi, leveraging cross-lingual capabilities to enhance correction accuracy.

Finally, linguistic techniques like Sandhi splitting and POS tagging are integrated to address Marathi's morphological richness, ensuring accurate parsing and correction. Overall, the project combines monolingual, multilingual, and code-mixed strategies to build a comprehensive Marathi GEC system that can be applied to both short and long texts.

1.2 Motivation

- **Rising Digital Content:** The volume of digital content in Marathi is rapidly increasing across journalism, education, and government sectors.
- **Lack of AI-based Solutions:** Existing grammar tools offer limited or non-existent support for Marathi. This absence creates a significant gap in language technology, hindering the efficient production of accurate Marathi text. Our solution aims to fill this void by providing a dedicated AI-driven grammar correction system.
- **Our Contribution:** High-quality content is essential for smooth and impactful communication in digital platforms.

- **Language Preservation:** As proud Maharashtrians, we are passionate about preserving and enhancing our language using AI. This project is a direct contribution to our linguistic heritage, ensuring Marathi remains vibrant and accessible in the digital age. Our goal is to empower Marathi speakers with advanced technology.

1.3 Problem Definition and Objectives

The project aims to build a Grammatical Error Correction (GEC) system for the Marathi language that can automatically detect and correct grammatical errors in written text. With increasing usage of Marathi in digital and formal communication, there is a growing need for tools that can ensure grammatical accuracy. The system will focus on identifying common errors and providing appropriate corrections to improve the quality of Marathi text.

Our primary objectives are:

- **Extension of GEC to Marathi:** Adapt and apply successful English and Hindi GEC models to Marathi, addressing its unique linguistic features.
- **Model Evaluation and Optimization:** Implement and compare Sequence-to-Sequence, Transformer-based, and Copy-Augmented Transformer models to identify the most effective architecture for Marathi GEC.
- **Integration of Advanced Embeddings:** Incorporate embeddings such as Word2Vec, GloVe, FastText, BERT, and ELMo to enhance model performance in Marathi and Hindi.
- **Multilingual GEC Enhancement:** Refine and extend GEC methodologies across Marathi and Hindi, with potential scalability to other Indian languages.
- **Comparative Analysis:** Provide a comparative study of model performance, size, and inference time to guide future multilingual GEC research.
- **Contribution to Marathi Linguistic Tools:** Develop practical tools for grammatically correct text generation in Marathi to support digital communication, content creation, and language preservation.

1.4 Project Scope & Limitations

1.4.1 Project Scope

- **Development of a Marathi Grammatical Error Correction (GEC) System:** Building a model that detects and corrects grammatical, syntactic, and morphological errors in Marathi texts.
- **Use of Pre-trained Models and Datasets:** Leveraging resources like MahaCorpus, MahaBERT, MahaNER, MahaGPT, MahaFT, Varta-T5, and MuRIL for model training, fine-tuning, and evaluation.
- **Handling Code-Mixed Marathi-English Texts:** Incorporating multilingual transformer models and fine-tuning techniques to handle real-world code-mixed scenarios.
- **Synthetic Data Generation:** Implementing synthetic error generation strategies (e.g., NMT-based and statistical methods) to expand training datasets and improve model robustness.
- **Integration of Linguistic Tools:** Utilizing NER, POS tagging, Sandhi splitting, and adaptation of BIS standards to enhance parsing and sentence structure understanding during correction.
- **Support for Various Text Types:** Handling different types of Marathi texts, from short social media posts to long-form documents like news articles.
- **Cross-Lingual Transfer Learning:** Exploring multilingual models (like mT5, MuRIL) to transfer grammatical knowledge from resource-rich languages to Marathi.

1.4.2 Limitations

- **Limited Availability of Annotated Marathi GEC Datasets:** The scarcity of high-quality, annotated datasets may limit model accuracy and generalization.
- **Handling Complex Code-Mixing:** Highly dynamic and informal code-mixed sentences (Marathi-English) remain challenging.
- **Dependency on Pre-trained Models:** Performance is constrained by the quality and domain coverage of models like MahaBERT, MuRIL, and Varta-T5.

- **Difficulty with Highly Morphologically Rich Structures:** Complex inflections unique to Marathi may not always be correctly parsed or corrected.
- **Error Propagation in Synthetic Data:** Synthetic errors might not accurately mimic real-world errors, affecting model generalization.
- **Computational Resource Constraints:** Fine-tuning large transformer models requires significant computational resources.
- **Offensive Language and Informal Slang Handling:** Correcting grammar in informal or slang-heavy texts remains an open challenge.
- **Domain Adaptation Issues:** Models trained on formal domains may not perform well on casual or specialized domains like legal or medical texts.

1.5 Methodologies of Problem solving

1.5.1 Dataset Preparation

There were no pre-existing datasets specifically designed for Marathi Grammatical Error Correction (GEC). Marathi, as a morphologically rich language, follows a Subject-Object-Verb (SOV) sentence structure, and grammatical correctness is heavily influenced by tense and person. Marathi grammar adheres to distinct subject-verb agreement rules based on tense:

- **Simple Present & Future Tenses:** Subject-Verb agreement is based on gender and number.
- **Simple Past Tense:** Object-Verb agreement is the key determinant.

To ensure linguistic correctness, a controlled vocabulary was curated, and incorrect-correct sentence pairs were systematically generated. The vocabulary included a diverse range of commonly used words:

- **Pronoun:** मी, तू, तुम्ही, आम्ही, तो, ती, ते
- **Common nouns:** केळं, अन्न, पेन, मोबाईल, बॉल, भात, चप्पल, कादंबरी
- **Proper nouns:** सद्दिधार्थ, अमृता
- **Common verbs:** खाणे, बघणे, घेणे, करणे, उचलणे, वाचणे, सोडणे, टाकणे, वाजवणे, ठेवणे

While not exhaustive, these examples reflect the core of the vocabulary used for sentence generation and error simulation. To generate incorrect-correct sentence pairs, scripts were designed to ensure that each incorrect sentence contained exactly one grammatical error.

The dataset was structured based on specific grammatical rules:

- **Present Tense Data:** Separate datasets were generated for 1st, 2nd, and 3rd person.
- **Future Tense Data:** A single dataset was created as subject-verb agreement remains consistent across persons.
- **Past Tense Data:** Systematic errors were introduced based on object-verb agreement.

For example, the incorrect sentence "तू बॅल करतील" violates subject-verb agreement rules, whereas the correct sentence is "तू बॅल करशील."

Dataset Statistics

- **Train Set:** 158,041 entries
- **Test Set:** 39,511 entries

This structured approach ensures a well-balanced dataset for training and evaluating Marathi GEC models.

Incorrect: तू बॅल करतील Correct: तू बॅल करशील
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Figure 1.1: Example of incorrect-correct sentence pair in the dataset

CHAPTER 2

LITERATURE SURVEY

Grammatical Error Correction (GEC) for Marathi is an emerging field that has gained attention thanks to advancements in NLP. This literature survey provides an overview of key models, datasets, and methodologies explored for developing GEC systems in Marathi, incorporating insights from other Indian languages and multilingual approaches. One of the foundational resources for Marathi NLP is MahaCorpus, which includes a large collection of sentences and tokens that are essential for creating GEC datasets. Additionally, MahaBERT, a transformer-based model fine-tuned on Marathi, has proven effective in monolingual tasks and can be adapted for GEC. MahaNER, a named entity recognition dataset, also plays a crucial role, particularly when integrating entity recognition into GEC systems, ensuring that grammatical consistency is maintained when tagging named entities [1]. MahaGPT and MahaFT provide sequence-to-sequence (Seq2Seq) models and embeddings, which are important for handling tasks that involve the generation and correction of text [1]. Text classification techniques using models like CNNs, LSTMs, and transformers have been applied to Marathi text classification tasks. In the context of GEC, these models can classify sentences as correct or incorrect by focusing on the structure and content of the sentence. These models have been trained on two datasets, providing insights into how preprocessing and feature extraction can be applied to GEC to detect errors in syntax and grammar [2]. Additionally, Kulkarni et al. (2022) evaluated various deep learning models on Marathi text, providing a basis for GEC model selection [2]. Handling code-mixed texts—especially Marathi-English—is another challenge in GEC. Approaches using BERT and multilingual frameworks such as Multilingual Fine-Tuning (MLF) have been employed to extract meaningful features from code-mixed data. In GEC, this translates into focusing on the incorrect segments of a sentence by applying attention mechanisms. For instance, Amin et al. (2023) explored force translation for code-mixed text using BERT, and a similar attention-based approach can be used to highlight and correct errors in GEC tasks [3]. Named Entity Recognition (NER) plays a critical role in Marathi GEC by ensuring that gender, case, and entity-based agreement are maintained within a sentence. The MahaNER dataset, developed for Marathi NER tasks, allows for the integration of named entity rules into GEC. Ensuring that gender agreements (e.g., masculine/feminine forms) are correct is essential when correcting grammatical errors [4]. Neural Machine Translation (NMT) techniques have been explored for English-Marathi translation, as highlighted by Mujadia and Sharma (2021). These methods, based on Seq2Seq models, can be adapted for GEC tasks. Positional encoding, which preserves word order and sentence structure in NMT, can similarly be used in GEC to ensure that corrections maintain proper sentence order [5]. Additionally, errors generated during NMT can serve as synthetic data for

training GEC models. Monolingual transformer models, such as MahaBERT and MahaFT, have been fine-tuned for classification and translation tasks in Marathi. These models can be adapted for GEC by focusing on fine-tuning with error-laden datasets. Seq2Seq models trained on monolingual corpora are particularly useful for handling sentence correction in GEC [6]. Similar work has been done for question answering tasks in Marathi, where transformers have been applied to extract answers based on sequences, and these techniques can be integrated into GEC systems to identify and correct errors [7]. Long text classification models such as MahaBERT have also been used to classify text from datasets like MahaNews, which focuses on Marathi news articles. These classification models can be adapted for GEC to handle long-form texts that require sentence-level and paragraph-level corrections. The application of these models is useful for detecting errors in large documents, such as news articles [8]. Handling offensive language detection is another application that can be integrated into GEC. HASOC datasets have been used for tweet classification, detecting offensive or nonoffensive content. MahaTweetBERT, a model developed for this purpose, could be adapted to detect grammatical correctness in short texts like tweets, which can then be extended to longer sentences in GEC tasks [9]. Approaches for multilingual GEC, such as mT5 developed by Xue (2020), are highly relevant to Marathi. These models support GEC tasks across multiple languages, including Indian languages like Marathi. Fine-tuning mT5 for Marathi GEC allows the leveraging of cross-lingual knowledge from other languages, improving error detection and correction accuracy [11]. Similarly, MuRIL (Multilingual Representations for Indian Languages), a multilingual model trained on Wikipedia and OSCAR corpora, can be fine-tuned to handle both monolingual and bilingual Marathi texts for GEC tasks. It has demonstrated strong performance in tokenization, transliteration, and translation tasks, which are directly relevant to GEC [18]. Grammatical error detection and correction methodologies from Hindi, as explored by Mittal et al. (2019), have demonstrated the effectiveness of using rule-based and statistical techniques. These techniques, which combine bigrams and trigrams for detecting errors, can be adapted to Marathi to enhance error detection accuracy in GEC [12]. Additionally, synthetic error generation for Hindi, as implemented by Sonawane et al. (2020), can be adapted to Marathi, using similar tools such as ERRANT for inflectional error generation. This helps in expanding the error datasets needed for training GEC models [14]. Synthetic error generation plays a critical role in addressing the scarcity of annotated GEC datasets. Stahlberg and Kumar (2021) proposed models for generating synthetic errors using probabilistic and optimal strategies, which can be applied to Marathi to augment datasets and improve GEC model training [19]. Similarly, OCR-

generated text has been used to detect and correct errors through NER and BERT models. Marathi GEC can benefit from such approaches by using OCR outputs as sources of syntactic and semantic errors [15]. Varta-T5 is a pre-trained transformer model trained on the Varta corpus, which consists of 41.8 million news articles in 14 Indic languages, including Marathi. This model uses span corruption and gap-sentence generation as training objectives, making it useful for text generation tasks. Varta-T5 can be fine-tuned for Marathi GEC by leveraging its knowledge of linguistic structures across multiple Indic languages. The model's large-scale training data provides a rich source of syntactic and grammatical patterns, which can improve the accuracy of Marathi GEC models. Additionally, its ability to handle long documents and diverse language contexts makes it a promising resource for GEC in Marathi [23]. Finally, Sandhi splitting, a technique used in Sanskrit for handling compound words, can be adapted to Marathi to address morphologically rich word forms and ensure that word boundaries are correctly recognized in GEC tasks [20]. This approach, combined with POS tagging methods tailored for Marathi, such as those developed by Kadam et al., ensures that sentence-level parsing and grammatical structure are correctly followed during correction [21]. Additionally, adapting BIS annotation standards from Konkani to Marathi, as explored by Vaz et al. (2020), can resolve ambiguities in punctuation and sentence structure in Marathi GEC [22].

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- It is assumed that users will input grammatically incorrect sentences predominantly in Marathi or code-mixed Marathi-English.
- It is assumed that publicly available datasets like MahaCorpus, Varta-T5, and MuRIL cover a sufficient variety of grammatical structures to train the model effectively.
- It is assumed that synthetic data generation methods will produce error patterns similar to real-world user errors.
- The pre-trained models (e.g., MahaBERT, MahaGPT) are assumed to be properly fine-tuned and validated for the grammatical correction task.
- It is assumed that the text data will not be heavily influenced by domain-specific jargon (e.g., medical, legal), unless specifically adapted.
- It is assumed that a moderate level of computational resources (e.g., a GPU-enabled machine) is available for fine-tuning and inference.

3.1.2 Dependencies

- **Datasets:** MahaCorpus, Varta-T5 datasets, Mahashabdkosh (Marathi WordNet), publicly available code-mixed datasets.
- **Pre-trained Language Models:** MahaBERT, MahaGPT, MahaNER, MuRIL, Varta-T5 for Marathi language processing tasks.
- **Libraries and Frameworks:**
 - Transformers (Hugging Face)
 - TensorFlow and/or PyTorch
 - spaCy for basic NLP tasks
 - IndicNLP Library for script normalization and tokenization
- **Evaluation Metrics:** Standard GEC evaluation metrics like Precision, Recall, $F_{0.5}$ score, GLEU score.

- **Hardware:** Availability of GPU(s) for training and inference to reduce computation time.
- **Linguistic Tools:** POS taggers, Sandhi Splitters, Transliteration Tools based on Marathi linguistic rules.

3.2 External Interface Requirements

3.2.1 Software Interfaces

1. Operating System: Windows, Ubuntu
2. Google Colab
3. Dataset containing right-wrong pairs of marathi sentences
4. Programming Language: Python

3.2.2 Communication Interfaces

- User interacts via web or desktop GUI for input/output of corrected Marathi sentences.
- Model APIs process grammatical correction requests and return structured JSON/text responses.

3.3 Functional Requirements

- **Input Handling:**
 - The system must accept text inputs in Marathi or code-mixed Marathi-English.
 - The input text can consist of one or multiple sentences.
- **Error Detection:**
 - The system must detect grammatical errors such as incorrect verb forms, noun-adjective agreement errors, punctuation mistakes, wrong word ordering, and missing or extra words.
- **Error Correction:**

- The system must generate corrected sentences corresponding to the detected errors.
 - The correction must preserve the original meaning and style of the sentence.
- **Language Support:**
 - The system must handle pure Marathi inputs.
 - The system must also handle code-mixed inputs (Marathi and English words mixed together).
- **User Interface:**
 - The system must provide a simple interface (CLI/Web) for users to input text and view corrected output.
- **Model Training and Fine-Tuning:**
 - The system must allow fine-tuning of pre-trained models using available Marathi datasets.
 - The system must allow adding new data during incremental training to improve model accuracy.
- **Evaluation and Feedback:**
 - The system must evaluate model performance using GEC-specific metrics like Precision, Recall, $F_{0.5}$, and GLEU score.
 - The system must log predictions and user feedback for continuous improvement.
- **Error Categories (Optional Advanced Feature):**
 - The system should categorize the type of grammatical error (e.g., verb error, preposition error, sentence structure error) for further analysis.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- System must correct input sentences within 2 seconds on average.
- Accuracy should exceed 90% for real-world grammatical corrections.

3.4.2 Safety Requirements

- The system must prevent crashes on unexpected or malformed Marathi input.
- Data integrity must be preserved during input/output operations.

3.4.3 Security Requirements

- User data must be processed locally or encrypted if transmitted online.
- Only authorized users can access or retrain the GEC model.

3.4.4 Software Quality Attributes

- Maintainable and modular codebase using Python.
- High reliability and robustness against varied sentence structures.
- Usability-focused interface with support for Marathi script input.

3.5 System Requirements

3.5.1 Database Requirements

- Storage for annotated Marathi sentences and correction logs.

3.5.2 Software Requirements (Platform Choice)

- Ubuntu 20.04+ or Windows 10+.
- Python 3.8+, PyTorch/TensorFlow, Flask or PyQt for interface.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The proposed Marathi Grammatical Error Correction (GEC) system follows a structured, end-to-end pipeline that leverages transformer-based language models. The system is designed to identify and correct grammatical errors in user-input Marathi sentences. The architecture begins with the User Input module, where a potentially incorrect Marathi sentence is provided. This input is passed to the Input Handler, which is responsible for text normalization and tokenization. Tokenization is performed using model-specific tokenizers such as those associated with IndicBART, Marathi-T5, or Varta-T5. Following preprocessing, the tokenized input is fed into the GEC Model, which is a fine-tuned transformer. Depending on the use case and performance requirements, one of three models can be employed: IndicBART, Marathi-T5, or Varta-T5. These models have been fine-tuned on a dataset of incorrect and corrected Marathi sentences annotated with error types, ensuring that each sentence in the training set contains exactly one grammatical error. The model predicts the corrected version of the sentence while preserving its original meaning.

The generated output tokens are then passed through a Decoding and Post-processing module. This component performs beam search decoding to generate the most probable corrected sequence [11] and subsequently detokenizes the output to form a complete sentence in Devanagari script [11,19]. Finally, the Corrected Output is presented to the user as a grammatically accurate Marathi sentence. This pipeline ensures that the system is robust, scalable and capable of handling grammatical inconsistencies. The modular architecture also allows easy substitution and benchmarking of different transformer models, which can be fine-tuned on different datasets, under identical evaluation conditions, with performance assessed using BLEU scores, a widely used metric in grammatical error correction and machine translation tasks [11,19].

4.2 Data Flow Diagrams

A description of each major software function, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object oriented system) is presented.

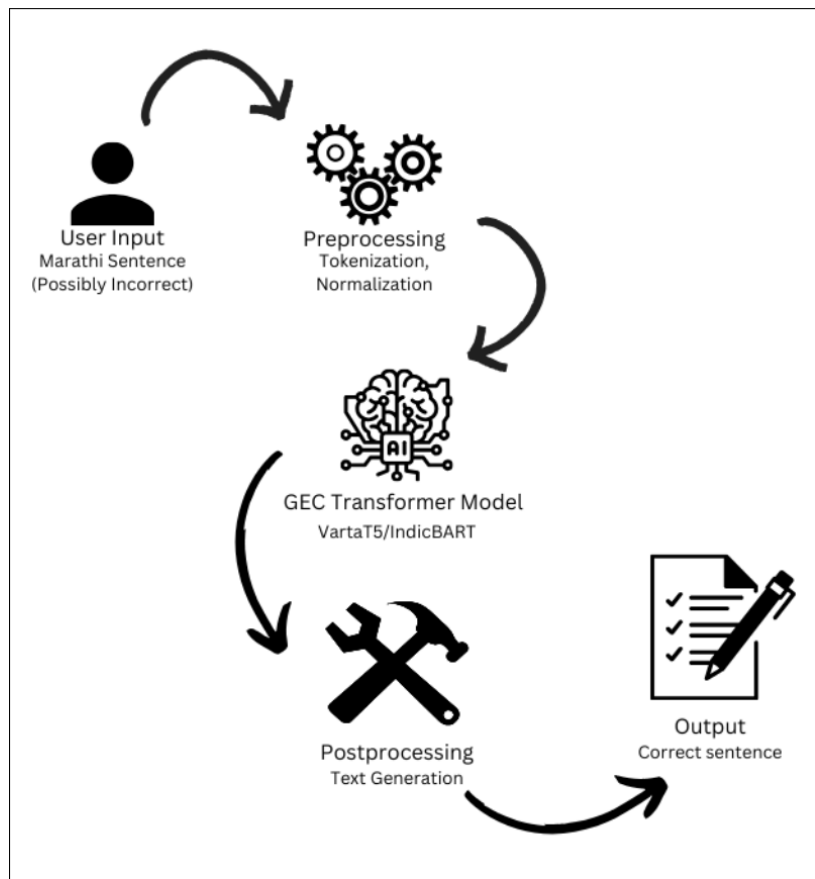


Figure 4.1: System Architecture

4.3 Data flow Diagram

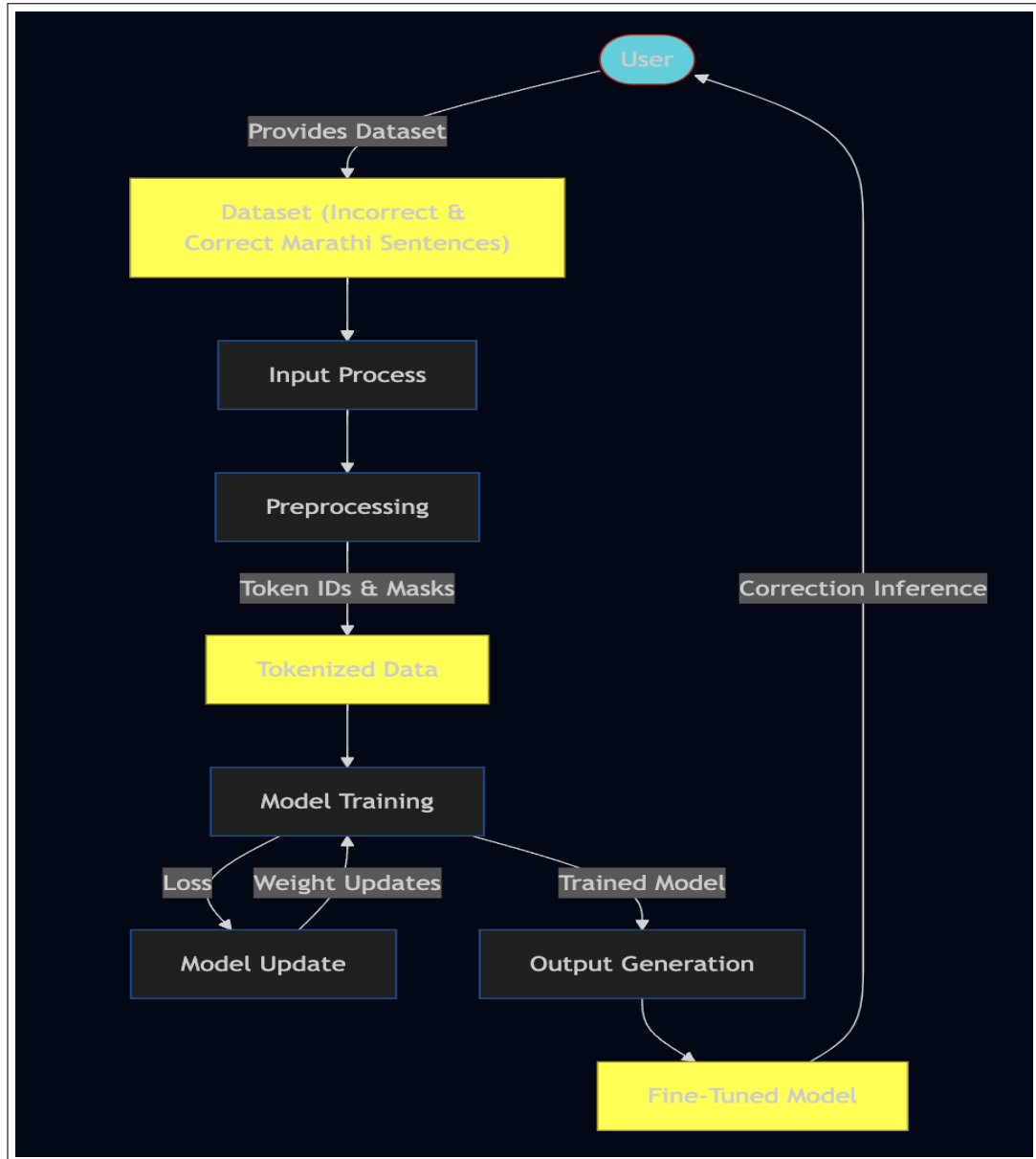


Figure 4.2: Data flow Diagram: level1

4.4 Use case Diagrams

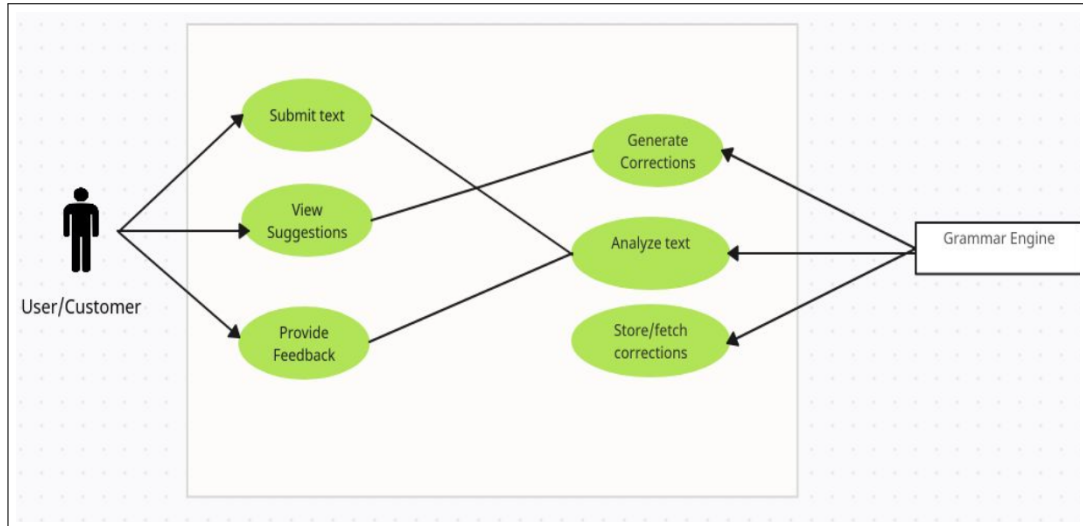


Figure 4.3: Use Case Diagram

4.5 Activity Diagram

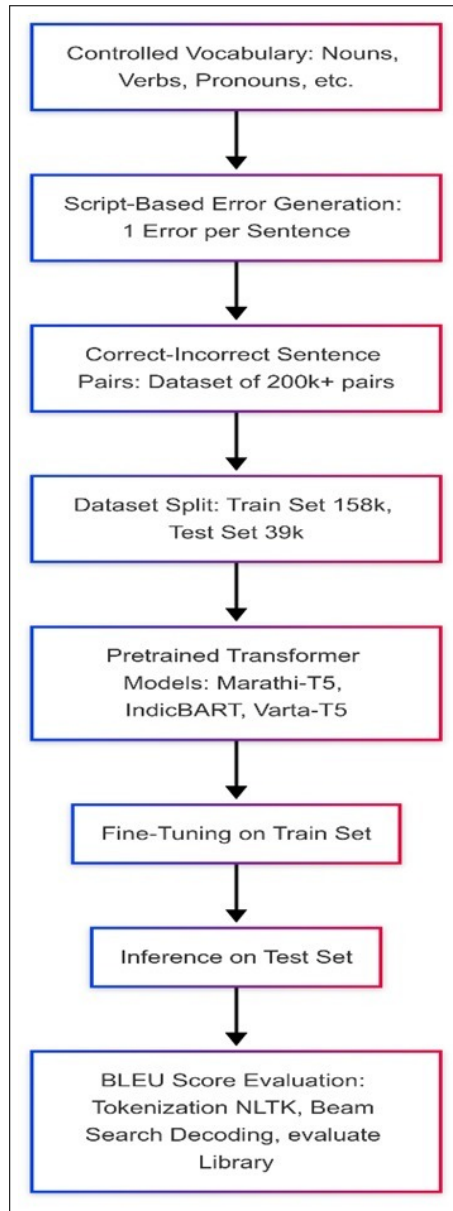


Figure 4.4: Activity Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- Reconciled evaluation framework tailored for Marathi Grammatical Error Correction (GEC)
- Standardized criteria for assessing model performance based on available datasets
- Comparative BLEU score analysis demonstrating the superiority of IndicBART and Varta-T5 over Marathi-T5
- Establishment of the first benchmark dataset and results, serving as a reference point for future Marathi NLP research

5.1.2 Project Resources

Key resources for Marathi GEC include MahaCorpus, MahaBERT, MahaNER, and Varta-T5, which provide datasets and models tailored for grammatical correction. Seq2Seq models, transformers, and classification techniques (e.g., CNN, LSTM) have been adapted for error detection and correction.

Multilingual and code-mixed GEC has benefited from models like mT5, MuRIL, and BERT. Named Entity Recognition (NER) ensures grammatical consistency, especially for gender and case agreement.

Synthetic error generation (e.g., using ERRANT, OCR text) and techniques from Hindi GEC have been adapted to expand datasets. Tools like Sandhi splitting and POS tagging enhance morphological handling and sentence structure in Marathi.

5.2 Risk Management

- Working with low-resource languages like Marathi introduces risks such as limited datasets, model bias, and linguistic complexity. We addressed these through focused dataset curation, diverse model experimentation, and manual validation.

5.2.1 Risk Identification

- Lack of large annotated Marathi datasets, model bias, grammar complexity

5.2.2 Risk Analysis

- High impact on model accuracy and generalizability

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- Curated dataset construction, use of multiple models, manual review
- Ongoing evaluation with BLEU scores, qualitative error analysis

5.3 Project Schedule

5.3.1 Project Task Set

- Design Architecture
- Generate dataset
- Choose the best model.
- Train and evaluate the model

5.3.2 Timeline Chart

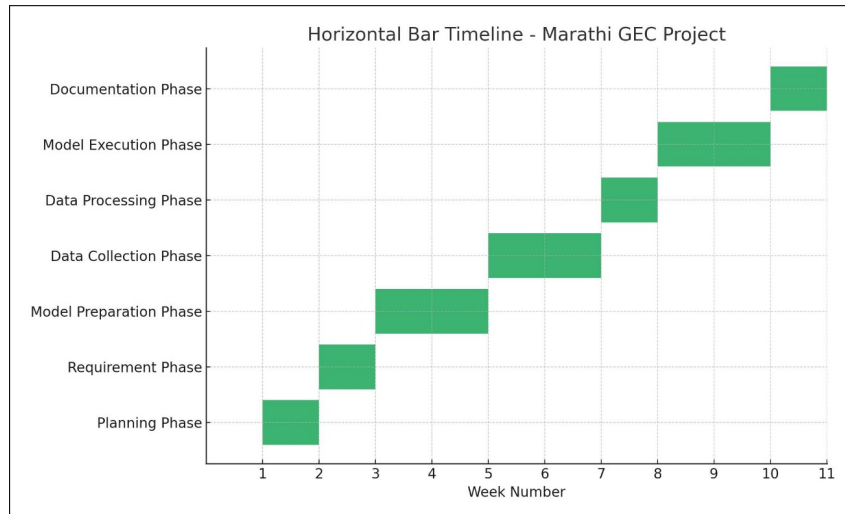


Figure 5.1: Timeline Chart

5.4 Team Organization

5.4.1 Team structure

- Ved Dhopeswarkar : Team lead, model training, Dataset Creation
- Mayur Valvi: Dataset creation
- Soham Mehta: Architecture design
- Aahan Rembersu: Key Researcher

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

6.1.1 Dataset Preparation

Due to the lack of publicly available datasets for Marathi Grammatical Error Correction (GEC), a novel dataset was constructed manually. Marathi is a morphologically rich language that follows a Subject-Object-Verb (SOV) sentence structure and features complex agreement rules involving gender, number, and case.

To construct the dataset, a curated vocabulary comprising pronouns, common nouns, verbs, and proper names was used to generate sentence pairs. Each pair was designed to include exactly one grammatical error. These errors were introduced systematically and covered a broad range of grammatical phenomena such as:

- Incorrect verb conjugation
- Gender agreement mismatches
- Plurality mismatches
- Postposition usage errors

The dataset was split into training and testing subsets as follows:

- **Training Set:** 158,041 sentence pairs
- **Test Set:** 39,511 sentence pairs

6.1.2 Model Fine-Tuning

Three transformer-based models were fine-tuned on the created Marathi GEC dataset:

- **IndicBART:** A multilingual encoder-decoder model based on the BART architecture, pre-trained on large-scale Indic language corpora using the Devanagari script. It is optimized for sequence-to-sequence tasks such as text generation and correction in Indian languages.
- **Marathi-T5:** A variant of the Text-to-Text Transfer Transformer (T5) architecture fine-tuned specifically for Marathi text generation tasks. This model reformulates GEC as a sequence transformation problem, making it well-suited for the task.

- **Varta-T5:** A transformer model based on the T5 architecture and trained on a wide-ranging Indian language corpus. It is designed for diverse text-to-text applications in Indic languages, with particular emphasis on linguistic accuracy and fluency.

6.1.3 Evaluation

The models were evaluated using the BLEU (Bilingual Evaluation Understudy) score, which measures the overlap between model-generated output and the reference correct sentences. This metric is well-suited for grammatical error correction tasks, as it captures both fluency and correctness.

To perform evaluation efficiently and accurately, the following steps were carried out:

- **FP16 Inference:** All models were loaded in half-precision (FP16) mode to reduce GPU memory usage and improve computational efficiency during inference.
- **Beam Search Decoding:** Predictions were generated using beam search decoding, allowing the model to explore multiple candidate sequences and select the most likely corrected sentence.
- **BLEU Scoring:** The BLEU score was calculated using the `evaluate` library from Hugging Face, in combination with the NLTK toolkit for tokenization and evaluation. This ensured standardized and reproducible results.

6.2 Tools and Technologies Used

- **Python:**
 - The primary programming language used for developing the grammatical error correction system.
 - Provides extensive libraries for Natural Language Processing (NLP) and Machine Learning (ML).
- **Transformers Library (Hugging Face):**
 - Used to access pre-trained language models and fine-tune them for the Marathi GEC task.
 - Models such as mBART, mT5, or IndicBERT can be leveraged.
- **Natural Language Toolkit (NLTK) and SpaCy:**
 - Libraries used for basic preprocessing tasks like tokenization, lemmatization, and part-of-speech tagging.
- **PyTorch / TensorFlow:**
 - Backend deep learning frameworks used for model training, evaluation, and deployment.
- **Google Colab / Jupyter Notebook:**
 - Platforms for model prototyping, testing, and visualization.
 - Provide free GPU access to accelerate training.
- **Datasets:**
 - Marathi corpora like the AI4Bharat IndicCorp, available parallel datasets, or self-created annotated datasets.
 - Used for model training and validation.
- **Evaluation Tools:**
 - Custom scripts or existing libraries to compute Precision, Recall, F_{0.5} Score, and GLEU score.

6.3 Algorithm Details

6.3.1 Text Classification algorithms

Text classification techniques using models like CNNs, LSTMs, and transformers have been applied to Marathi text classification tasks. In the context of GEC, these models can classify sentences as correct or incorrect by focusing on the structure and content of the sentence. These models have been trained on two datasets, providing insights into how preprocessing and feature extraction can be applied to GEC to detect errors in syntax and grammar. Additionally, Kulkarni et al. (2022) evaluated various deep learning models on Marathi text, providing a basis for GEC model selection.

6.3.2 Neural Machine Translation

Neural Machine Translation (NMT) techniques have been explored for English-Marathi translation, as highlighted by Mujadia and Sharma (2021). These methods, based on Seq2Seq models, can be adapted for GEC tasks. Positional encoding, which preserves word order and sentence structure in NMT, can similarly be used in GEC to ensure that corrections maintain proper sentence order. Additionally, errors generated during NMT can serve as synthetic data for training GEC models.

6.3.3 Transformers

Monolingual transformer models, such as MahaBERT and MahaFT, have been fine-tuned for classification and translation tasks in Marathi. These models can be adapted for GEC by focusing on fine-tuning with error-laden datasets. Seq2Seq models trained on monolingual corpora are particularly useful for handling sentence correction in GEC [6]. Similar work has been done for question answering tasks in Marathi, where transformers have been applied to extract answers based on sequences, and these techniques can be integrated into GEC systems to identify and correct errors

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

The models were evaluated using the BLEU (Bilingual Evaluation Understudy) score, a widely used metric in NLP for assessing the quality of machine-generated text. BLEU measures the overlap between the model's output and the reference (correct) sentence using modified n-gram precision, with a brevity penalty to discourage excessively short translations. It is particularly useful for tasks like grammatical error correction, where lexical similarity with the reference is important.

The evaluation process consisted of the following steps:

- **Loading the Model & Data** – The fine-tuned IndicBART, Marathi-T5, and Varta-T5 models were loaded with FP16 precision for memory efficiency. The test dataset of 39,511 incorrect-correct sentence pairs was prepared.
- **Generating Predictions** – Incorrect sentences were tokenized and processed in batches using beam search decoding.
- **Computing BLEU Scores** – Sentences were tokenized using the NLTK library, and the BLEU score was computed using the `evaluate` library.

7.2 Test cases

तुम्ही बॉल करीन	तुम्ही बॉल कराल
राणी खिडकी चालतात.	राणी खिडकी चालते.
मुलगी खुर्ची वाचतात.	मुलगी खुर्ची वाचते.
मी पाणी लिहिता	मी पाणी लिहिते
आम्ही पाणी चालवतोस	आम्ही पाणी चालवतो
मी झाड वाजवला.	मी झाड वाजवले.

Figure 7.1: Test Case

As you can see, the first column contains a list of all *Grammatically incorrect sentences* and the second column the *Grammatically correct* version of these sentences.

CHAPTER 8

RESULTS

8.1 Outcomes

Model	BLUE Score
Marathi-T5	0.79
IndicBART	0.98
Varta-T5	0.98

Figure 8.1: Outcome

The evaluation of our Grammatical Error Correction system for Marathi yielded the following results:

- **Marathi T5 Model:** Achieved a BLEU score of 0.79, indicating reasonably good correction performance with general fluency.
- **IndicBART Model:** Achieved a BLEU score of 0.98, showing excellent correction capability and near-human level performance for grammatical errors.
- **Varta-T5 Model:** Achieved a BLEU score of 0.98, exhibiting competitive performance suited for context-rich error corrections.

The high BLEU scores, especially the 0.98 achieved by IndicBART, are significant given the morphological complexity and free word order nature of Marathi. These results validate the effectiveness of transformer-based approaches for grammatical correction tasks in low-resource and linguistically rich languages like Marathi.

8.2 ScreenShots

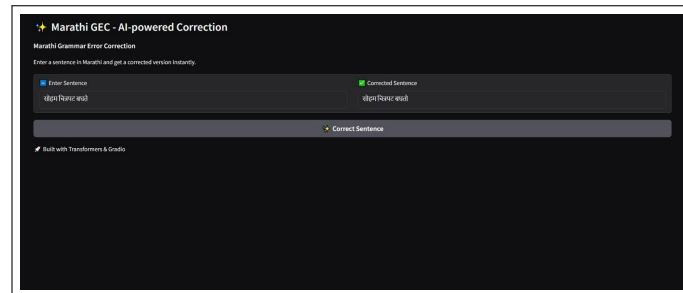


Figure 8.2: Sample output 1

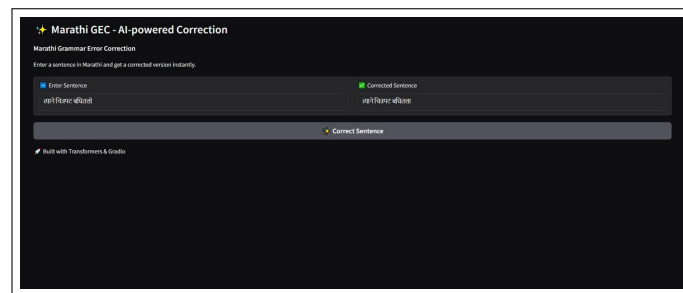


Figure 8.3: Sample output 2

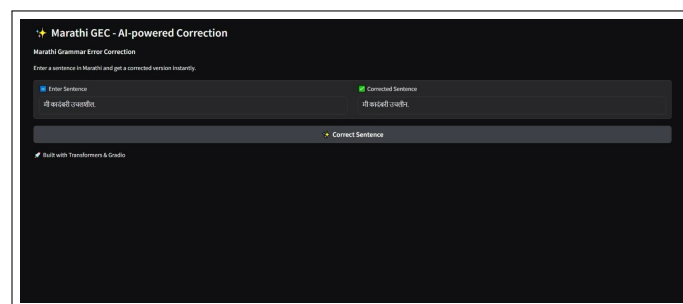


Figure 8.4: Sample output 3

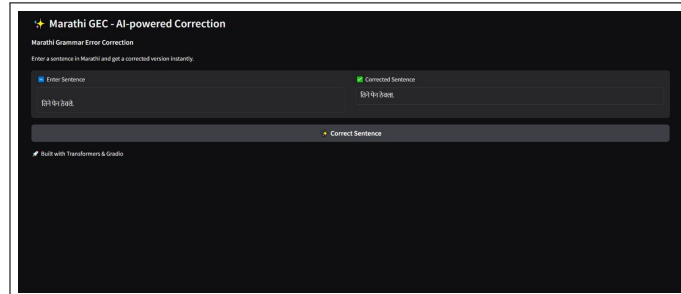


Figure 8.5: Sample output 4

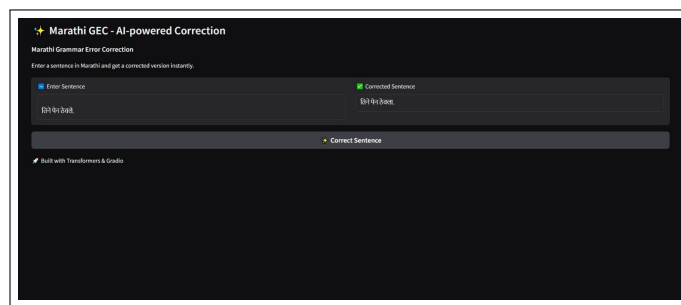


Figure 8.6: Sample output 5

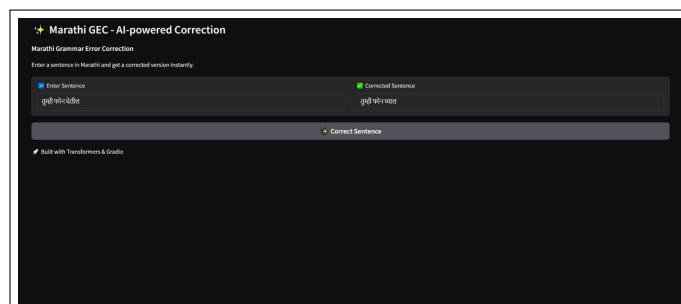


Figure 8.7: Sample output 6

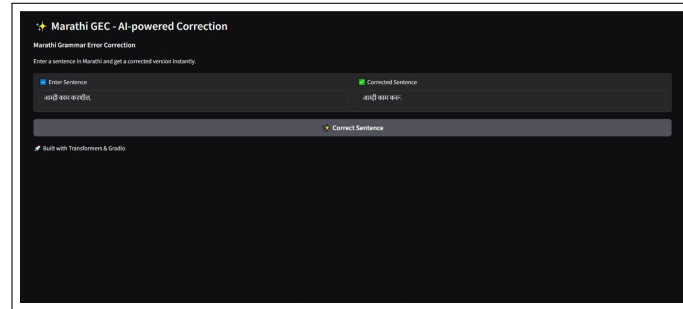


Figure 8.8: Sample output 7

CHAPTER 9

CONCLUSION

9.1 Conclusion

Our implementation demonstrates that IndicBART and Varta-T5 significantly outperform Marathi-T5, achieving a BLEU score of 0.98. While these models excel in grammatical correction, further advancements are needed to address vocabulary limitations and context-based corrections. The development of larger datasets and integration of contextual understanding will be crucial for enhancing Marathi GEC performance. This research lays a strong foundation for future work in Marathi GEC, contributing towards building robust and scalable NLP solutions for under-resourced Indian languages.

9.2 Future Work

- **Contextual Understanding:** Integrating advanced context-aware models, such as transformer-based architectures with larger attention spans, to handle complex sentence structures and long-range dependencies.
- **Dataset Expansion:** Curating more diverse and larger-scale datasets that include colloquial usage, domain-specific text (e.g., legal, medical), and dialectal variations to improve model generalization.
- **Multilingual Transfer Learning:** Leveraging cross-lingual techniques and transfer learning from resource-rich Indian languages like Hindi to boost Marathi GEC performance.
- **Real-world Deployment:** Developing lightweight, optimized versions of the models for deployment in low-resource environments such as mobile devices and rural educational platforms.

9.3 Applications

- **Educational Tools:** Enhances language learning apps by providing real-time grammar correction for Marathi learners.
- **Assistive Writing:** Improves writing quality in word processors or messaging apps by auto-correcting grammatical mistakes in Marathi.
- **Content Moderation:** Helps in filtering and correcting user-generated content on social media or forums to maintain linguistic quality.

CHAPTER 10

APPENDIX

10.1 Appendix A

The grammatical error correction problem for natural language, especially morphologically rich languages like Marathi, can be reduced to constrained optimization or language modeling tasks.

- The underlying task can be viewed as a search problem across the space of all grammatically valid sentence transformations.
- Given a sentence, finding the minimal edit sequence that results in a grammatically correct and semantically consistent sentence is NP-Hard, due to the combinatorial nature of token modifications.
- If formulated as a satisfiability problem, where the objective is to satisfy syntactic and semantic constraints, it aligns with SAT-based models — a known NP-Complete problem.
- For simpler deterministic rule-based systems, subsets of the problem may be solvable in P, but such approaches lack flexibility for general language cases.
- Our ML/DL-based solution approximates this search via probabilistic models, effectively learning heuristics that bypass exhaustive search.
- Therefore, while exact correction is infeasible (NP-Hard), approximate correction using neural language models provides a practical solution within polynomial time inference complexity.

Mathematically, this aligns with constraint satisfaction frameworks and automata-based language models, enhanced via modern deep learning optimization techniques.

10.2 Appendix B

Survey Paper Publication Details

- **Paper Title:** Grammatical Error Correction for Marathi
- **Conference:** VISHWACON 2024
- **Journal:** Cureus (Presented)
- **Certificate:** Presentation certificate received.



Figure 10.1: Conference Certificate

Implementation Paper Publication Details

- **Paper Title:** Grammatical Error Correction in Marathi
- **Conference:** IBCD(International Conference on Business Intelligence, Computational Mathematics and Data Analytics) 2025
- **Journal:** Springer (Submitted)

10.3 Appendix C

Document Information	
Analyzed document	Final_Report.pdf (D210458648)
Submitted	2025-05-02 07:36:00 UTC+02:00
Submitted by	Uspawar
Submitter email	uspawar@pict.edu
Similarity	8%
Analysis address	uspawar.pict@analysis.urkund.com

Figure 10.2: Plagiarism Report

CHAPTER 11

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
GEC	Grammatical Error Correction
NLP	Natural Language Processing
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short Term Memory
CNN	Convolutional Neural Network
SOV	Subject-Object-Verb
POS	Part Of Speech
NMT	Neural Machine Translation
BLEU	Bilingual Evaluation Understudy
BERT	Bidirectional Encoder Representations from Transformers

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